



CBA: Backdoor Attack on Deep Speech Classification via Audio Compression

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Abstract

As deep neural networks permeate various aspects of society, research has demonstrated their vulnerability to backdoor attacks, particularly with untrustworthy third-party platforms, enabling attackers to manipulate model outputs using customized triggers. However, most audio-based backdoor methods are either sample-agnostic or audible. In this paper, we propose **Compression-based Backdoor Attack (CBA)**, a novel strategy designed specifically for audio compression scenarios, which also utilizes fusion as well as iterative optimization strategies to generate sample-specific triggers and ensure both the effectiveness and stealthiness of the triggers. Extensive experiments validate the attack efficiency and robustness.

Index Terms: Backdoor attack, deep learning, audio compression, speech classification.

1. Introduction

Over the past several years, Deep Neural Networks (DNNs) have evolved continuously and gained widespread recognition for their superior performance, permeating various aspects of daily life, including privacy and security-sensitive areas. Ensuring the security of DNNs is paramount, yet, despite their superior performance, they are vulnerable across various stages, from data collection to model deployment. And these vulnerabilities can be exploited by malicious attacks. One such vulnerability is the risk of backdoor attacks [1].

The efficacy of DNNs is highly dependent on extensive training datasets and considerable computational resources, imposing a significant burden on individual researchers and small teams. To avoid the labor-intensive process of creating datasets, researchers often resort to third-party datasets or data sourced on the Web, which can potentially compromise both data accuracy and security. Furthermore, training large-scale DNNs requires substantial hardware resources, prompting researchers to consider cost-saving solutions, such as outsourcing. Two common approaches are Machine Learning as a Service (MLaaS) and transfer learning, both of which would introduce security concerns, including the threat of backdoor attacks.

Backdoor attacks aim to embed hidden threats within a model that are triggered under specific conditions to manipulate the model's output. In the audio domain, Koffas et al. [2] utilized ultrasonic as triggers. Liu et al. [3] selected corresponding triggers from a predefined pool of natural or ambient audible sounds. Cai et al [4]. proposed adding a small portion of audio as a trigger using the auditory masking effect. However, some of these methods are audible and even inaudible ones severely disrupt the audio spectrogram. In addition, other methods [5], [6], [7] proposed stealthy backdoor attacks, but require higher poisoning ratio for effective attack, making them more suscep-

tible to detection. Considering that additive noise is more easily noticeable and detectable by humans, meanwhile, achieving a balance between attack effectiveness and stealthiness presents a significant challenge, this paper introduces a novel method named **CBA** that generates backdoor audio samples by compressing the original audio, which results in imperceptible and sample-specific triggers that closely resemble the original audio. Our contributions are as follows:

- We introduce a backdoor attack via audio compression to inconspicuously poison the training data and can be passively activated.
- We designed an iterative optimization algorithm to generate sample-adaptive triggers while maintaining the stealthy of the triggers. With just a 5% injection rate of poisoning samples, we achieve an attack success rate exceeding 99%.
- We conduct extensive experiments to compare our method with existing approaches, and evaluate its performance in terms of effectiveness, stealth and robustness.

2. PRELIMINARIES

2.1. Problem Formulation

A speech classification model can be formulated as a function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ represents the input feature domain and $\mathcal{Y}$ is a set of k categories. The parameter θ denotes the model learned from the training dataset $D_t = \{(x_i, y_i) | x_i \in \mathcal{X}, y_i \in \mathcal{Y}\}$. A modified poisoned sample (x_i^*, y_t) is generated by trigger embedding function $G(\cdot)$. The poisoned dataset can be generated by injecting triggers into a subset of the clean dataset D_{sub} , i.e. $D_{poi} = \{(x_i^*, y_t) | x_i^* = G(x_i), (x_i, y_i) \in D_{sub}, y_t \in \mathcal{Y}\}$, where y_t is the target label specified by the attacker. The remaining clean samples compose clean dataset D_c . Poisoning ratio is defined as $\gamma = |D_{poi}|/|D_t|$, where a lower γ indicates a more efficiency attack.

During training, x_i is transformed to $T(x_i)$ by the 2D time-frequency transform function $T(\cdot)$. The training of a clean model is formulated as a single-level optimization problem:

$$\theta = \arg \min_{\theta} \sum_{(x_i, y_i) \in D_t} \mathcal{L}(f_\theta(T(x_i)), y_i), \quad (1)$$

where $\mathcal{L}$ denotes the loss function, e.g. cross-entropy loss.

For the backdoored model training, the model $f_{\theta'}$ is considered successful only if it behaves normally on clean samples and outputs the specified target label on poisoned samples:

$$\begin{cases} f_{\theta'}(T(x_i)) = y_i, \\ f_{\theta'}(T(x_i^*)) = y_t. \end{cases} \quad (2)$$

Thus, it becomes a bi-level optimization problem:

$$\begin{aligned}\theta' &= \arg \min_{\theta'} \mathcal{L}(D_c \cup D_{poi}, f_{\theta'}) \\ &= \arg \min_{\theta'} \sum \mathcal{L}(f_{\theta'}(T(x_i), y_i) + \sum \mathcal{L}(f_{\theta'}(T(x_i^*), y_i)).\end{aligned}\quad (3)$$

2.2. Threat Model

2.2.1. Attackers' Knowledge and Capabilities

Our attack is designed for MLaaS scenario, in which the victim uploads a validated dataset and model to a third-party platform for training, and then hosts the trained model on platform and provides API interface for users. In this scenario, attackers can generate poisoned samples by accessing training data but cannot obtain any information about the model structure.

2.2.2. Attackers' Goal

The attacker's objectives can be summarized in the following three goals:

- **Effectiveness:** The primary goal is to embed a backdoor into the model while maintaining its original performance and achieving a high attack success rate.
- **Stealthiness:** Poisoned samples must remain sufficiently stealthy, with minimal differences from clean samples.
- **Robustness:** We define robustness in two aspects: (1) the embedded backdoor should resist classic defense methods; (2) poisoned samples should preserve trigger features and remain effective after data preprocessing.

3. PROPOSED METHOD

In this study, we highlight the distinctive properties of audio compression that make it particularly suitable for backdoor attacks. Audio compression is widely used to reduce data size while maintaining perceptual quality. The process is inherently content-dependent, leading to variable perturbations across samples. Furthermore, compression can also be seen as a data enhancement method that improves the performance of the model and counteracts overfitting by introducing possible perturbations. Given these properties, we propose an audio compression-based trigger mechanism for backdoor attack as illustrated in Fig. 1.

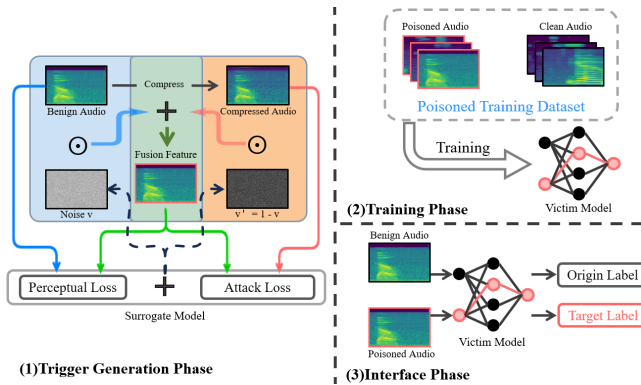


Figure 1: Overall pipeline of CBA.

3.1. Trigger Generation with Sample Interpolation

The methodology employed in our research comprises two principal components: audio compression and interpolation masks generation. Initially, the audio data is compressed using two distinct techniques: bit compression and wavelet compression.

- **Bit Compression:** We scale the audio to predetermined range of bit depths followed by integer quantization.
- **Wavelet Compression:** Wavelet decomposition is utilized to segregate the audio signal's frequency content by multilayer wavelet decomposition, after which the resulting coefficients are compressed.

The compressed samples are used as the initial backdoor samples. However, they differ significantly from the original audio, as shown in Fig. 2, making them easily detectable by the victim. Therefore, to enhance the stealthiness of the triggers within the compressed audio samples, i.e., to increase the similarity between the backdoor samples and the original samples, we propose a method to embed the features of the compressed audio into the features of the original audio. This method effectively reduces the unnatural perturbations introduced by the compression process. We achieve this by designing a mask v_i that controls the interpolation of the compressed features. Our objective is to maintain the effectiveness of the compressed features as triggers while minimizing the difference between the feature of compressed audio and original audio.

The feature fusion process is defined as follows:

$$T(x_i^*) = T(x_i') \odot v_i + T(x_i) \odot (1 - v_i), \quad (4)$$

where x_i^* , x_i' , x_i represents the poisoned, compressed and original sample respectively, $\odot$ denotes element-wise multiplication.

The mask v_i is generated through an iterative optimization process, guided by a loss function that measures the discrepancy between the compressed and original audio features. In the aforementioned black-box scenario, structure of the victim model is not known to the attacker, so this loss is computed via a surrogate model θ^* . In this work, Resnet18 [8] is adopted as the surrogate model since it is simple and the training cost is trivial. The loss function comprises two components: attack loss and perceptual loss.

We obtain sample x_i^t of the target category that most similar to the sample to be poisoned based on cosine similarity and optimize the poisoned sample along this direction to improve optimization efficiency. The attack loss can be calculated as:

$$\mathcal{L}_{att} = \frac{\|f_{\theta^*}(T(x_i^*)) - f_{\theta^*}(T(x_i^t))\|_2}{\|f_{\theta^*}(T(x_i^t))\|_2}. \quad (5)$$

The following perceptual loss aims to smooth x_i^* and make it similar to the original sample x_i , and it can be defined as:

$$\begin{aligned}\mathcal{L}_{per} &= \frac{\|f_{\theta^*}(T(x_i^*)) - f_{\theta^*}(T(x_i))\|_2}{\|f_{\theta^*}(T(x_i))\|_2} \\ &+ 10 \cdot \log_{10} \left(\frac{\sum (T(x_i))^2}{\sum (T(x_i) - T(x_i^*))^2} \right).\end{aligned}\quad (6)$$

The final loss function is a combination of the attack and perceptual losses:

$$\min \mathcal{L} = \lambda \mathcal{L}_{att} + \mathcal{L}_{per}, \quad (7)$$

The detailed generation procedure is illustrated in Algorithm 1.

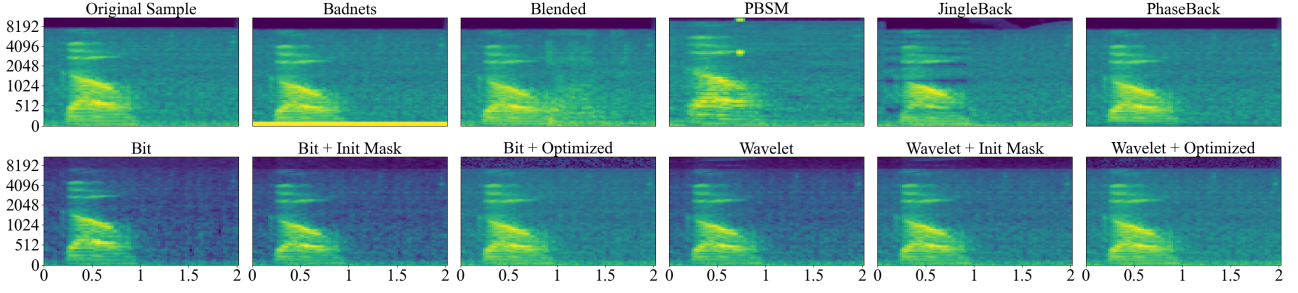


Figure 2: Visualization of the results. As shown above, BadNets and PhaseBack introduce anomalies in the lower part of the spectrogram, while other three baselines cause global perturbations across the entire spectrogram.

Algorithm 1: Poisoned Training Data Generation

Input: Surrogate model θ^* , iter epoch E , learning rate η , mask elements range $[lb, ub]$, target label y_t .
Output: Poisoned dataset D_{poi} .

```

1 for each sample  $(x_i, y_i) \in D_{sub}$  do
2   Compress audio:  $x'_i = \text{Compress}(x_i)$ 
3   Compute max cosine similarity  $\max_{\cos}(x_i, x'_i)$ ;
4   for  $j = 1, 2, \dots, E$  do
5     Compute loss:  $\mathcal{L} = \lambda \cdot \mathcal{L}_{att} + \mathcal{L}_{per}$ ;
6     Calculate gradient  $g_i^j$ ;
7     Update mask  $v_i = \text{clip}(v_i - \eta \cdot g_i^j, lb, ub)$ ;
8   end
9   Generate poisoned sample by (4)
10  Change label:  $y_i \rightarrow y_t$ 
11 end
12 return Poisoned dataset  $D_{poi}$ ;
```

4. EXPERIMENTS

4.1. Experimental Setup

Dataset and Models. This paper adopts Google Speech Commands dataset [9]. From this dataset, 21,115 audio samples with 10 labels were selected. We evaluated the performance using three deep learning models, ResNet34 [8], LSTM [10], and KWT [11].

Attack Setup. We use Resnet18 as the surrogate model and train for only 10 epochs with 40% training data. Each mask v_i was optimized 200 iterations, with a learning rate η of 0.0002, and mask value bounds lb, ub of 0.9 and 0.2. We set λ to 2.5 as default and the poisoning ratio was set to 5% following [12]. The attack results were averaged over three independent experiments.

Training Setup. All experiments were conducted using the PyTorch framework on Nvidia RTX A6000 GPUs. We set the batch size of the victim model to 256. Both models use Adam optimizer with a learning rate of 0.002, and a cross-entropy loss function.

Baseline and Evaluation Metrics. We compare CBA with several baseline methods, including BadNets [13], Blended [14], Ultrasonic [15], PBSM [4] and PhaseBack [7]. Two metrics are used for evaluation: (1) **Accuracy of Clean Data (ACC)**: measures the percentage of correctly classified audio samples without triggers. (2) **Attack Success Rate (ASR)**: indicates the percentage of poisoned samples classified as target labels by the victim model.

4.2. Main Results

Table 1: BA (%) AND ASR (%) OF DIFFERENT ATTACKS

Method	ResNet34		LSTM		KWT	
	ACC	ASR	ACC	ASR	ACC	ASR
Vanilla	94.56	-	92.39	-	91.33	-
BadNets	94.80	98.51	92.85	95.92	92.36	97.08
Blended	96.08	97.86	92.95	<u>98.52</u>	92.16	98.61
JingleBack	95.69	97.67	91.38	94.56	91.67	92.62
PBSM	95.59	<u>99.68</u>	<u>93.54</u>	98.25	91.96	99.43
PhaseBack	94.80	96.91	92.45	93.60	91.96	87.47
Bit-r	95.60	98.92	93.63	96.70	90.11	98.96
Wavelet-r	95.01	98.74	92.56	94.57	91.87	93.35
Bit	96.69	99.73	92.36	98.57	92.46	<u>99.02</u>
Wavelet	<u>96.48</u>	99.09	92.36	95.40	92.86	97.79

The best results are highlighted in **bold**, the second-best results are underlined. The bottom four lines show our proposed method, where 'r' indicates the use of a random mask and bottom two lines represents a mask based on iterative optimization.

As shown in Table 1, CBA outperforms existing methods in both ASR and ACC. Experiments show that CBA-Bit achieves 99.73% ASR and 96.69% ACC on ResNet34, which significantly outperforms traditional methods. The iterative optimization strategy further improves the attack effectiveness. Although slightly outperformed by PBSM on the KWT, our method is superior in stealthiness, in contrast, PBSM introduces noticeable sound and artifacts in spectrogram. In addition, CBA excels in cross-model robustness, demonstrating strong model architectures adaptability.

4.3. Resistance to Potential Defenses

4.3.1. Resistance to Filtering

Table 2: Resistance results of CBA against Filtering

Method	Metrics	Lowpass	Gaussian	Median	Wiener
Bit	ACC	93.300	95.988	95.247	95.325
	ASR	99.565	99.783	99.957	92.957
Wavelet	ACC	94.040	95.559	96.182	95.286
	ASR	99.652	99.478	99.043	99.043

As show in Table 2, we evaluated the impact of various filtering methods on ResNet34. Low-pass filtering reduced the

ACC but had no effect on the ASR. For other filtering methods, there is no significant impact on ASR, demonstrating both stronger robustness and high resistance to filter-based defenses.

4.3.2. Pruning [16]

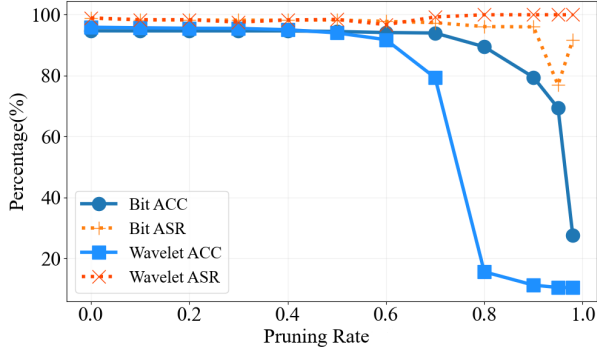


Figure 3: Resistance results of CBA against pruning.

Fig. 3 shows the ACC and ASR versus the number of neurons pruned. The results indicate that ACC drops much faster than ASR as more filters are pruned. This suggests that there is no pruning point that can reliably mute the backdoor before the model becomes inoperable.

4.3.3. Fine-Tuning [17]

Table 3: Resistance results of CBA against Fine-Tuning

Method	Metrics	Before FT	After FT
Bit	ACC	94.720	93.728
	ASR	98.912	95.624
Wavelet	ACC	95.910	93.845
	ASR	98.957	93.922

We fine-tuned the poisoned model using 20% of clean data, and the results in Table 3 showed a decline in both ACC and ASR. However, the backdoor could not be effectively removed.

4.3.4. STRIP [18]

Fig. 4 shows the detection results of applying STRIP to test samples for the model poisoned by CBA. It demonstrates that the clean and backdoor samples have similar and non-separable entropy distributions. Since the triggers of CBA are sample-specific, superimposing different content onto the poisoned samples cannot destroy corresponding triggers.

4.3.5. Neural Cleanse [19]

The anomaly index of the reverse-engineered trigger identified by Neural Cleanse for the actual target class in the model poisoned by CBA-Bit and CBA-Wavelet is 1.86 and 1.74 respectively, which falls below the threshold of 2. This indicates that CBA successfully evades Neural Cleanse detection. The reason is that Neural Cleanse is specifically designed to detect tiny and fixed triggers, making it incapable of reverse-engineering an adaptive, global, and sample-specific backdoor trigger.

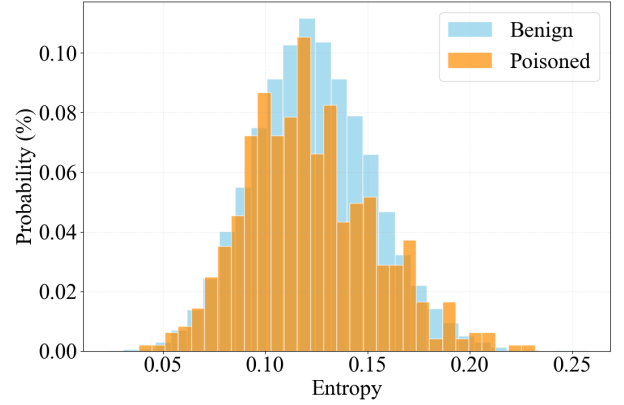


Figure 4: Resistance results of CBA against STRIP.

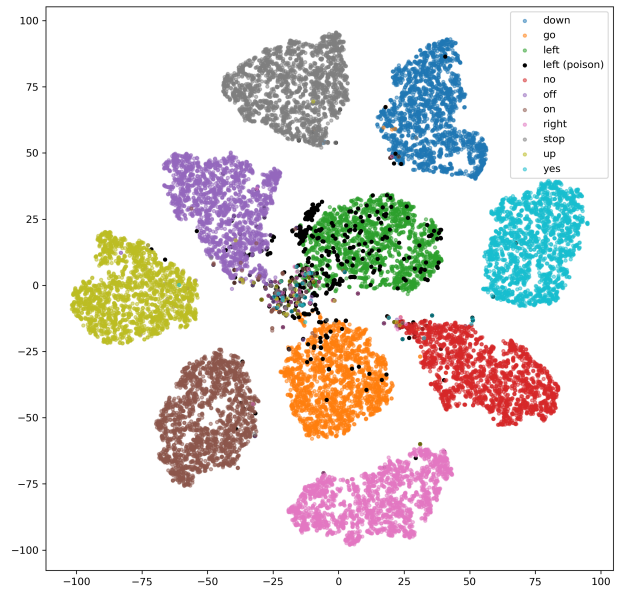


Figure 5: Visualization of training samples via T-SNE. The majority the poisoned samples generated by CBA are uniformly spread across the data manifold of the target class.

4.3.6. T-SNE [20]

We further visualize the activation values of high-dimensional features in Fig. 5. The results demonstrate that most of poisoned samples generated by CBA are evenly distributed along the data manifold of the target class, indicating that separating poisoned samples based on activation values is infeasible.

5. CONCLUSION

This letter proposes an effective inaudible backdoor attack paradigm for speech classification. The proposed attack is validated on three deep learning models, demonstrating high attack success rates without affecting the prediction accuracy of normal inputs across all tested models. Our method surpasses existing backdoor attacks in both stealth and effectiveness, while also exhibiting robust against several widely used and powerful backdoor defense countermeasures.

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